

Retrofit Governance Framework

Preventing cross-market code divergence in GEB / FCC

UOB Global Electronic Banking | Wipro | Version 1.0 | 20 August 2026

Context. On 18 August 2026 the bank raised an escalation following the OFM MY and HK code review, citing inconsistent implementation of the same retrofit across markets. The specific finding was that the company ID and company abbreviation validation in MY uses a different boolean condition from SG and HK, producing different pass and fail behaviour for the same input scenarios. The bank's stated position is that a retrofit is like-for-like unless a deviation is documented and approved.

Purpose of this document. To set out how retrofit work across the seven GEB markets is tracked, applied, verified and signed off, so that divergence is prevented at source and coverage is provable at every release.

Scope. All code changes applied to more than one country codebase: CR delivery, defect and SIT fixes, production hotfixes, security and CVE remediation, and platform or dependency changes. Markets in scope: SG, MY, CN, TH, ID, OV, HK.

Constraints this design respects. Legacy Ant based build, no modern CI pipeline assumed, Bitbucket and Jira as the only workflow tooling, seven long lived country branches, and a delivery calendar that cannot absorb a tooling programme. Nothing here requires a build change, a new licence, or a change to the JBoss / Turbine / Torque stack.

1. Failure modes we are designing against

These are the mechanisms that actually produce divergence. Each control later in this document maps back to one or more of them.

ID	Failure mode	How it shows up	Control
F1	Invisible fan-out	One base change implies up to seven country changes, but only the base change is tracked. A partially applied retrofit is indistinguishable from a complete one.	C1, C2
F2	Stale delta	A change is retrofitted everywhere, then corrected during SIT in one market. The correction reaches some markets and not others. The earliest markets keep the superseded logic.	C1, C6
F3	Manual re-implementation	The change is read and retyped in the next branch rather than applied as a diff. Boolean operators, negation placement and condition order drift. Same intent, different behaviour. This is the OFM case.	C3, C4
F4	No cross-market reviewer	Each PR is reviewed inside its own country branch. No single reviewer ever compares all versions, so nobody owns the equivalence question.	C4, C5
F5	Untracked change	Work delivered without a Jira, or with a Jira raised only in one market. Nothing to fan out from and nothing to audit against later.	C1, C3
F6	Undocumented deviation	A market genuinely needs different logic, but the reason lives in a developer's head or a chat thread. At review time it is indistinguishable from a mistake.	C1, C5

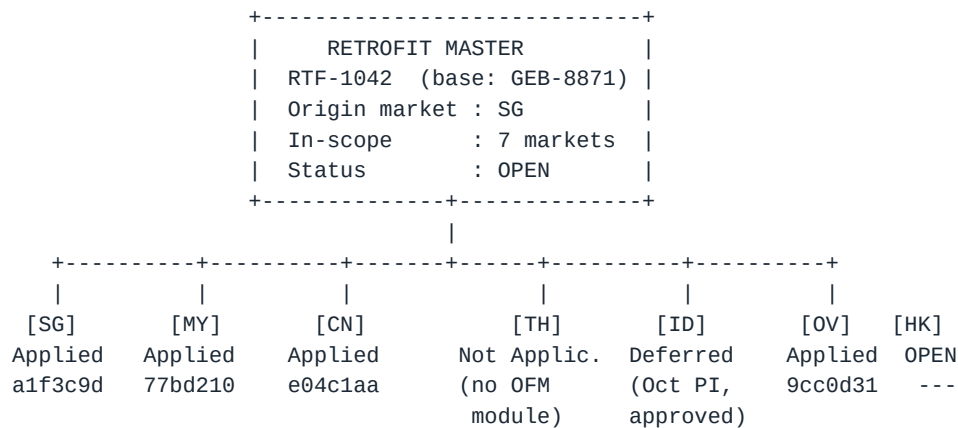
Note on volume. Seven markets, concurrent PIs, hotfix and CVE tracks running in parallel is a genuinely high fan-out. That is the reason to install a control, not an explanation for the misses. Any process that depends on every developer remembering every downstream market will fail at this volume.

2. Design principles

1. **Absence must be a decision, not a gap.** Every market is either done, explicitly not applicable with a reason, or explicitly deferred with a target. There is no silent state.
2. **Carry the diff, not the description.** Retrofits are applied as patches from the source branch. Re-typing is the defect mechanism, so remove it.
3. **Verification is mechanical, not attentional.** A script compares the result across branches. Human review confirms intent; it does not carry the burden of spotting a single inverted operator.
4. **The tracker is the truth, not a status deck.** Coverage is derived from Jira and commit IDs, never maintained by hand in a spreadsheet that drifts from reality.
5. **The parent stays open until the last market is live.** A retrofit is not complete when the first country ships.
6. **Evidence is produced as a by-product of doing the work,** not assembled afterwards for an audit.

3. Target operating model

The unit of control is the **Retrofit Master**: one ticket representing a change that must exist in more than one market, with one **Country Retrofit item** beneath it per in-scope market, created up front.



Master cannot close while any child is OPEN or DEFERRED.

3.1 Country item states

State	Meaning	Mandatory evidence
Open	Not yet applied to this market.	None. Blocks master closure and blocks that market's release sign-off.
Applied	Change is in the country branch and equivalent to the origin.	Commit ID or IDs, PR link, equivalence report result.
Applied with deviation	Change is in, but deliberately differs from the origin.	Commit ID, written reason, named approver, equivalence report showing the intended difference only.
Not applicable	The market does not carry the affected module, product or flow.	Written reason naming the module or product that is absent, and who confirmed it.
Deferred	Applicable but consciously scheduled later.	Target PI or release, approver, and a linked follow-up item.

3.2 Jira implementation

Two options depending on how quickly the bank's Jira administration can move. Option A is the target; Option B delivers the same control with zero schema change and can start this week.

Element	Option A: schema change	Option B: no schema change
Master	New issue type <i>Retrofit Master</i>	Existing Task or Story with label <i>RETROFIT-MASTER</i>
Country item	Sub-task type <i>Country Retrofit</i>	Sub-task with label <i>RTF-<market></i>
Linkage	Native parent and child	Parent and child, plus link type <i>relates to</i> the base CR or defect
Market	Custom field <i>Market</i> , single select	Component or label per market
Outcome	Custom field <i>Retrofit Outcome</i> with the five states	Resolution plus a mandatory comment template

Element	Option A: schema change	Option B: no schema change
Enforcement	Workflow validator: master cannot transition to Done while any child is not resolved	Board query plus release checklist. Manual but visible

3.3 The retrofit board

A Jira Kanban board scoped to retrofit work only. Configuration:

- **Board filter:**

```
project = GEB AND (labels in (RETROFIT-MASTER, RETROFIT)
    OR issuetype in ("Retrofit Master", "Country Retrofit"))
ORDER BY Rank ASC
```

- **Swimlanes:** one per Retrofit Master. The whole fan-out for a change sits on one row, so an incomplete fan-out is visible at a glance rather than inferred.
- **Columns:** To Apply → In Progress → In Review → Applied → Verified. Verified means the equivalence report is attached and reviewed, not merely that the PR merged.
- **Quick filters:** per market (SG, MY, CN, TH, ID, OV, HK), *Deviations*, *Deferred*, *Origin market only*, and *Aged over one PI*.
- **Aged item query**, reviewed weekly:

```
labels = RETROFIT AND status not in (Verified, Done)
AND created <= -30d ORDER BY created ASC
```

3.4 Coverage matrix

The board shows flow. The matrix shows completeness, and it is the artifact that goes into release sign-off and to the bank. It is generated from the Jira filter export, never typed by hand.

Retrofit	Base	SG	MY	CN	TH	ID	OV	HK	State
RTF-1042	GEB-8871	a1f3c9	77bd21	e04c1a	N/A	DEF	9cc0d3	OPEN	Blocked
RTF-1043	GEB-8902	b7e104	c39aa1	11d4f8	6ba22c	ff01e3	2c7d91	40ab6e	Complete
RTF-1044	IDB-151301	N/A	DEV	N/A	N/A	d90c11	N/A	N/A	Deviation

Legend: commit hash = applied and verified; **N/A** = not applicable with recorded reason; **DEF** = deferred with approved target; **DEV** = applied with approved deviation; **OPEN** = outstanding. A row is Complete only when no cell is OPEN.

4. Bitbucket discipline

4.1 Commit standard

Every commit that forms part of a retrofit carries the market, the country item and the origin, so the fan-out is reconstructable from history alone even if Jira is later reorganised.

[<COUNTRY-ITEM>] [<MARKET>] retrofit of <ORIGIN-JIRA> (<ORIGIN-COMMIT>) : <summary>

Example:

[GEB-8871-MY] [MY] retrofit of GEB-8871 (a1f3c9d) : OFM company id and abbrev validation

Deviation commits must additionally carry:

Deviation: <reason> | Approved-by: <name>

4.2 Propagation method

The change is carried as a patch. The developer does not retype it. Where the country branches sit in one repository:

```
git checkout version-38.1_J8_MY
git cherry-pick -x a1f3c9d          # -x records the origin commit in the message
# resolve conflicts, but only where the file genuinely differs
git diff a1f3c9d^ a1f3c9d          # origin diff
git diff HEAD^ HEAD                # what was actually applied here
# the two diffs must be identical apart from recorded deviations
```

Where the markets are separate repositories:

```
git format-patch -1 a1f3c9d -o /tmp/rtf1042
cd ../geb-my && git apply --3way /tmp/rtf1042/0001-*.patch
```

The final step, comparing the origin diff with the applied diff, is the whole point. It is a thirty second check that closes failure mode F3 completely.

4.3 Pull request requirements

- A retrofit PR must reference its country item ID in the title, and must not be raised without one.
- The PR description carries the equivalence report output, or a link to it on the Jira.
- Reviewers: the normal country reviewer, plus the **designated cross-market reviewer**. The second reviewer's job is not to review the code, it is to confirm that this version matches the other markets and that any difference is a recorded deviation.
- A retrofit PR does not merge while its origin change is still open in SIT in another market. This is the guard against propagating a version that is about to be corrected (F2).

4.4 Enforcement options

Which of these applies depends on the Bitbucket deployment:

Deployment	Available enforcement
Bitbucket Data Center / Server	Pre-receive hook or the Yet Another Commit Checker plugin to reject commits that do not match the message pattern. Merge checks to require the second reviewer's approval on branches carrying retrofit PRs.
Bitbucket Cloud	No server side hooks. Enforce through the PR template, the required reviewer list, and the weekly audit query. Weaker, but the audit still catches non-compliance within a week.

If neither is available quickly, the control still holds through the Jira country items and the release gate. The commit standard makes the audit cheap; it is not the primary control.

5. Mechanical equivalence check

A small script, run by the developer before raising the PR and re-run by the cross-market reviewer. It answers one question: for the files this retrofit touched, are the country branches the same?

5.1 What it does

1. Takes a retrofit ID, the list of touched file paths, the origin branch and the in-scope market branches.
2. Extracts each file from each branch with `git show <branch>:<path>`.
3. Normalises: strips trailing whitespace, collapses indentation, removes blank lines and comment-only lines. Formatting differences are noise and must not drown the signal.
4. Compares each market against the origin and classifies the result as **identical**, **formatting only**, or **semantic difference**.
5. For semantic differences, prints the differing lines side by side and, where the file is Java, names the enclosing method by scanning back to the nearest signature.
6. Writes a report suitable for attaching to the Jira and pasting into the PR.

5.2 Illustrative output

```
RETROFIT EQUIVALENCE REPORT   RTF-1042   origin: SG (version-38.1_J8_SG)
file: com/misys/portal/ofm/service/OFMGroupValidator.java

MY version-38.1_J8_MY   *** SEMANTIC DIFFERENCE ***   method: validateGroup()
SG : if (!(...equals(companyId) || !currentUser.getCompany_abbrev_name()...
MY : if (!(...equals(companyId) && currentUser.getCompany_abbrev_name()...
CN version-38.1_J8_CN   identical
TH version-38.1_J8_TH   formatting only (2 lines, indentation)
HK version-38.1_J8_HK   identical
ID version-38.1_J8_ID   FILE ABSENT -> confirm Not Applicable
OV version-38.1_J8_OV   identical

RESULT: 1 semantic difference, 1 absence to confirm.  NOT CLEAR TO MERGE.
```

5.3 Honest limits

- It compares the files a retrofit declares it touched. If a retrofit is applied to a different file in another market, the tool will not know to look there. The country item's file list is what makes this work.
- It cannot tell an intentional market difference from a mistake. That is why deviations are recorded on the Jira, and the reviewer reconciles the report against the recorded list.
- Legitimate divergence already exists across markets from years of market-specific development. First runs will produce noise. The answer is to baseline per file: the first run records known differences, and the check

thereafter reports only new ones.

Build effort is on the order of two to three days for a working version, in Python, run from a developer laptop against a local clone. No CI dependency, no build system change. If a scheduler exists it can be wired in later to run nightly across all open retrofits.

6. Release gate

Coverage becomes an exit criterion. For each country release:

Check	Evidence	Owner
No open retrofit cells for this market	Coverage matrix export	Country SPOC
Every retrofit in the drop has an equivalence report attached	Jira attachments	Cross-market reviewer
All deviations for this market are listed and approved	Deviation register	Solution Architect
Deferred items carry an approved target PI	Country item fields	Delivery lead
Test scope names every market the change lands in	Test scope document	Developer and test lead

The last line addresses a known recurring gap. Where a change has multiple market variants or sub-flows, the handover to testing must state all of them explicitly rather than relying on release comments.

7. Special cases

Case	Rule
Defect fix on an already retrofitted change	The Retrofit Master reopens automatically. Country items are recreated for every market that took the superseded version. This is the F2 control and the single most important rule in this document.
Production hotfix with no Jira	A retrofit item is raised retrospectively within twenty four hours of the fix going in, with the commit ID attached. Emergency speed is preserved; the audit trail is not lost.
Change genuinely specific to one market	Still gets a Retrofit Master, with the other six markets closed as Not Applicable and a reason. This costs two minutes and removes the ambiguity that currently makes every single-market change look like a possible miss.
Approved deviation	Recorded on the country item before the code is written, with the approver named. It then appears in the deviation register and is expected by the equivalence check rather than flagged by it.
Long-lived market divergence discovered during a retrofit	Logged as a finding on the Retrofit Master and raised separately. Do not silently align it inside an unrelated retrofit, and do not silently leave it.

8. Roles

Role	Accountable for
Developer	Applying the patch rather than retyping, running the equivalence check, attaching the report and commit IDs, and stating every affected market at test handover.

Role	Accountable for
Cross-market reviewer	The equivalence call across all markets. Reviews the report, not just the country PR. Named, and with a named backup so the control does not stall on one person's leave.
Country SPOC	The state of that market's column at release. Confirms Not Applicable claims for their market.
Solution Architect	Approving deviations and owning the deviation register.
Delivery lead	Chairing the weekly retrofit review, clearing aged items, approving deferrals.
Programme lead	The coverage matrix, the release evidence pack, and reporting to the bank.

9. Cadence

- **Weekly, fifteen minutes:** retrofit review on the board. Open cells, aged items, deferrals. This is the only recurring meeting cost of the framework.
- **Per release:** coverage matrix and evidence pack produced as part of the release checklist.
- **Per PI:** a coverage back-scan sampling closed retrofits to confirm the matrix reflects the code, not just the tickets.

10. Metrics

Metric	Definition	Target
Retrofit coverage	Country items resolved divided by total, per release	100% at sign-off
Escapes	Inconsistencies found by the bank or in production, not by us	Zero
Self-detection rate	Inconsistencies found by our own check as a share of all found	Above 90%
Fan-out lead time	Days from origin merge to last market applied	Within the same PI
Undocumented deviations	Differences found with no approval record	Zero

11. Rollout

Phase	Window	Deliverable
0. Contain	Week 1	Align the MY OFM validation to SG and HK for the MY Sep PI. Confirm no other market carries the same OFM divergence. Circulate the commit standard and PR checklist.
1. Back-scan	Weeks 1 to 3	Reconstruct the last two PIs of retrofits across all seven markets. Publish the first coverage matrix with every gap found, and a remediation plan per gap. Findings go to the bank proactively.
2. Structure	Weeks 2 to 4	Retrofit Master and country items live in Jira, board configured, all new retrofits raised through it. Cross-market reviewer and backup named.
3. Verify	Weeks 3 to 6	Equivalence script working, baselines captured for high fan-out files, mandatory on all new retrofits.

Phase	Window	Deliverable
4. Gate	From the next release	Coverage matrix and evidence pack become a formal release exit criterion. Offered to the bank as a standing release artifact.

12. Risks to this plan

Risk	Mitigation
Framework is followed for a month and then quietly abandoned under delivery pressure	The release gate is the only thing that keeps it alive. If coverage is not an exit criterion, this becomes a document. Everything else is optional; the gate is not.
Back-scan surfaces a large number of historical inconsistencies	Expected, and better found by us. Classify by risk, remediate high risk in the current PI, schedule the rest, and present the full picture rather than a filtered one.
Equivalence check produces too much noise and gets ignored	Baseline known differences on first run and report only new divergence. A check that cries wolf is worse than no check.
Single reviewer becomes a bottleneck	Named backup, and the review is of a generated report rather than of raw code, so it takes minutes.
Bank reads this as process theatre	Lead with the back-scan findings, not with the framework. Evidence of gaps we found ourselves is the credibility, not the document.

Prepared by Sai Pavan Jaldu, Wipro, GEB engagement. For internal review before any bank-facing circulation.